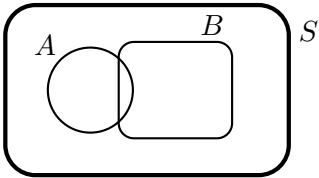


1. Let $A = \{12n \mid n \in \mathbb{Z}\}$, and let $B = \{k \in \mathbb{Z} \mid k \text{ is even}\}$, and let $C = \{3j \mid j \in \mathbb{Z}\}$. Show that $A \subseteq B \cap C$. That is, show that if $x \in A$, then $x \in B$ and $x \in C$.

Let’s start with some notation and visual intuition. Generally speaking we’ll be drawing an example as below, with A and B as subsets of a larger “universe” S .



Operation	Expression	Picture	Description
Union	$A \cup B$		$x \in A \cup B$ if $x \in A$ or $x \in B$.
Intersection	$A \cap B$		$x \in A \cap B$ if $x \in A$ and $x \in B$.
Set difference	$A \setminus B$		$x \in A \setminus B$ if $x \in A$ and $x \notin B$.
Complement	A' (or A^c or $\overline{A}$)		$x \in A'$ if $x \notin A$.
Subset	$A \subseteq B$ (or $B \supseteq A$)		$x \in A \implies x \in B$.

For each of the following, determine whether the statement is true for all sets A , B and C . If an equality fails, determine whether one or the other of the possible inclusions holds. Prove your assertions.

2. $(A \cap B) \cup (A \setminus B) = A$

3. $A \cup (B \setminus C) = (A \cup B) \setminus (A \cup C)$

4. Suppose $A = \{k \in \mathbb{Z} \mid 3|k\}$ and $B = \{2k \mid k \in \mathbb{Z}\}$ are subsets of $\mathbb{Z}$. Describe (a) $(A \cup B)'$ and (b) $(A \cap B)'$.